Intro to limits

Intuitive idea of limits

Please read the following articles on Khan Academy

- Limits intro
- Estimating limit values from graphs
- Using tables to approximate limit values

and complete the following practice

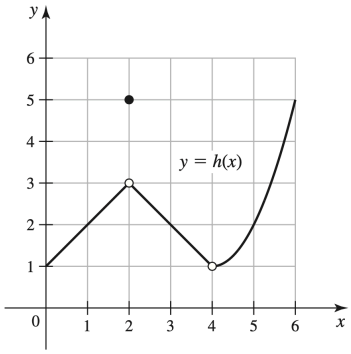
- Limits intro
- Estimating limit values from graphs
- Creating tables for approximating limits
- Estimating limits from tables

Remark. Khan Academy at various points uses the meaningless phrase "infinitely close"; what they mean to say is "arbitrarily close".

1.) Use the graphs below to estimate the given values or to state that they do not exist:

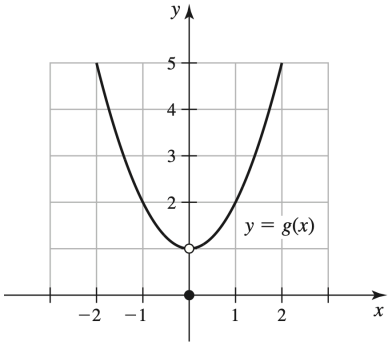
Finding limits from a graph Use the graph of h in the figure to find the following values or state that they do not exist.

a. $h(2)$ **b.** $\lim_{x\to 2} h(x)$ **c.** $h(4)$ **d.** $\lim_{x\to 4} h(x)$ **e.** $\lim_{x\to 5} h(x)$



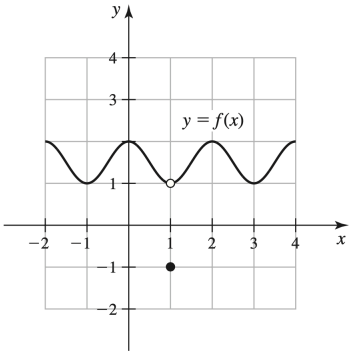
Finding limits from a graph Use the graph of g in the figure to find the following values or state that they do not exist.

a. $g(0)$ **b.** $\lim_{x\to 0} g(x)$ **c.** $g(1)$ **d.** $\lim_{x\to 1} g(x)$



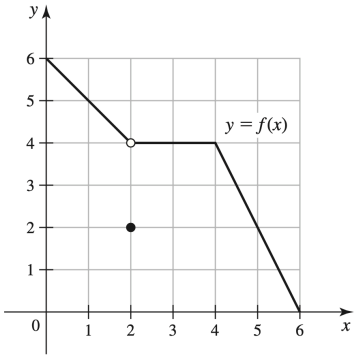
Finding limits from a graph Use the graph of f in the figure to find the following values or state that they do not exist.

a. $f(1)$ **b.** $\lim_{x\to 1} f(x)$ **c.** $f(0)$ **d.** $\lim_{x\to 0} f(x)$



Finding limits from a graph Use the graph of f in the figure to find the following values or state that they do not exist.

a. $f(2)$ **b.** $\lim_{x\to 2} f(x)$ **c.** $\lim_{x\to 4} f(x)$ **d.** $\lim_{x\to 5} f(x)$



2.) Use the tables below to estimate the following limits:

(a) $\lim_{x\to 2} \frac{(x^2-4)}{x-2}$

(b) $\lim_{x\to 1} \frac{(x^3-1)}{x-1}$

(c) $\lim_{x\to 1} \frac{x-9}{\sqrt{x}-3}$

(d) What mathematical constant does $\lim_{x\to 1}(1+x)^{\frac{1}{x}}$ appear to approach?

$$f(x) = \frac{(x^2-4)}{x-2}$$

x	$f(x)$
1.9	3.9
1.99	3.99
1.999	3.999
1.9999	3.9999
2.0001	4.0001
2.001	4.001
2.01	4.01
2.1	4.1

$$f(x) = \frac{(x^3-1)}{x-1}$$

x	$f(x)$
0.9	2.71
0.99	2.9701
0.999	2.997001
0.9999	2.9997
1.0001	3.0003
1.001	3.003001
1.01	3.0301
1.1	3.31

$$f(x) = \frac{x-9}{x^{\frac{1}{2}}-3}$$

x	$f(x)$
8.9	5.9832868
8.99	5.9983329
8.999	5.9998333
8.9999	5.9999833
9.0001	6.0000167
9.001	6.0001667
9.01	6.0016662
9.1	6.0166206

$$f(x) = (1+x)^{\frac{1}{x}}$$

x	$f(x)$
0.01	2.7048138
0.001	2.7169239
0.0001	2.7181459
0.00001	2.7182682
−0.00001	2.7182954
−0.0001	2.7184178
−0.001	2.7196422
−0.01	2.731999